

AegisBrain

AI-Powered Industrial Knowledge Intelligence Platform
ET AI Hackathon 2026 - Track 8

The Problem: Knowledge Fragmentation

- **Industrial engineers waste 35% of their time searching for information.**
 - Average large plants operate 7 to 12 disconnected document systems.
 - Fragmentation contributes to 18-22% of unplanned downtime.
 - 25% of experienced engineers will retire soon, taking undocumented knowledge.
 - Our Goal: Unify scattered PDFs, manuals, and sensor data into one intelligent platform.

The Solution: AegisBrain

- **A premium, AI-powered operations dashboard for heavy industry.**
 - Universal Document Ingestion: Upload PDFs & extract equipment tags instantly.
 - Interactive Knowledge Graph: Visually map equipment to sensors, rules, and work orders
 - Expert QA Copilot: Chat with your manuals using an integrated RAG engine.
 - Safety & Compliance Auditor: Autonomous AI agent cross-references active permits.

Target Audience & Business Viability

- **Field Technicians:** Get instant, mobile-friendly answers on the factory floor.
 - Safety Officers: Proactively detect compliance violations before accidents happen.
 - Reliability Engineers: Accelerate Root Cause Analysis with the Knowledge Graph.
 - Plant Managers: Reduce unplanned downtime and scale across the enterprise.
 - Impact: Translates directly into millions of dollars saved in reduced downtime.

Technology Stack & Architecture

- **100% Client-Side Architecture for ultra-fast, zero-latency demos.**
 - UI Framework: Semantic HTML5, Vanilla CSS (Industrial Theme), Vanilla JS.
 - Knowledge Graph: D3.js force-directed mapping.
 - Document Intelligence: PDF.js for real-time, in-browser client-side extraction.
 - AI Engine: Dual-mode RAG (Offline Simulator + Live Google Gemini API Integration).

Evaluation Criteria Proven

- **Entity Extraction: Accurately identifies tags like V-202 and OISD-150.**
 - Query Quality: Copilot delivers domain-expert answers with exact citations.
 - Graph Completeness: 20 interconnected nodes mapping full asset relationships.
 - Speed: 2-second RAG response vs 45-minute manual search.
 - Compliance Accuracy: AI flags sub-standard wall thickness and active gas leaks.

Live Demonstration

Switching to AegisBrain Dashboard...